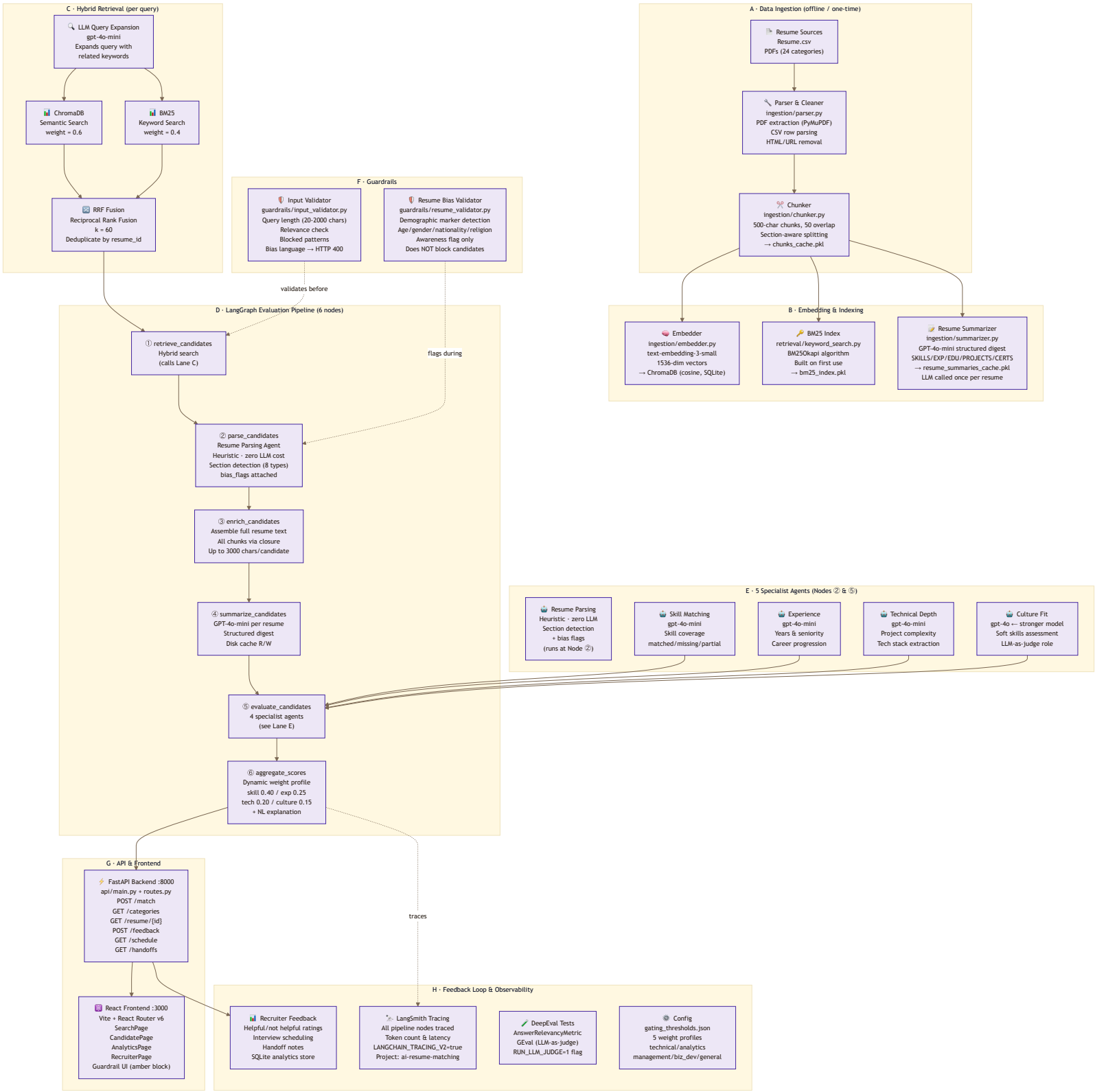
C · Hybrid Retrieval (per query)
🔍 LLM Query Expansion
gpt-4o-mini
Expands query with
related keywords
📊 ChromaDB
Semantic Search
weight = 0.6
📊 BM25
Keyword Search
weight = 0.4
🔀 RRF Fusion
Reciprocal Rank Fusion
k = 60
Deduplicate by resume_id
F · Guardrails
🛡️ Input Validator
guardrails/input_validator.py
Query length (20-2000 chars)
Relevance check
Blocked patterns
Bias language → HTTP 400
🛡️ Resume Bias Validator
guardrails/resume_validator.py
Demographic marker detection
Age/gender/nationality/religion
Awareness flag only
Does NOT block candidates
validates before
flags during
A · Data Ingestion (offline / one-time)
📄 Resume Sources
Resume.csv
PDFs (24 categories)
🔧 Parser & Cleaner
ingestion/parser.py
PDF extraction (PyMuPDF)
CSV row parsing
HTML/URL removal
✂️ Chunker
ingestion/chunker.py
500-char chunks, 50 overlap
Section-aware splitting
→ chunks_cache.pkl
B · Embedding & Indexing
🧠 Embedder
ingestion/embedder.py
text-embedding-3-small
1536-dim vectors
→ ChromaDB (cosine, SQLite)
🔑 BM25 Index
retrieval/keyword_search.py
BM25Okapi algorithm
Built on first use
→ bm25_index.pkl
📝 Resume Summarizer
ingestion/summarizer.py
GPT-4o-mini structured digest
SKILLS/EXP/EDU/PROJECTS/CERTS
→ resume_summaries_cache.pkl
LLM called once per resume
D · LangGraph Evaluation Pipeline (6 nodes)
① retrieve_candidates
Hybrid search
(calls Lane C)
② parse_candidates
Resume Parsing Agent
Heuristic · zero LLM cost
Section detection (8 types)
bias_flags attached
③ enrich_candidates
Assemble full resume text
All chunks via closure
Up to 3000 chars/candidate
④ summarize_candidates
GPT-4o-mini per resume
Structured digest
Disk cache R/W
⑤ evaluate_candidates
4 specialist agents
(see Lane E)
⑥ aggregate_scores
Dynamic weight profile
skill 0.40 / exp 0.25
tech 0.20 / culture 0.15
+ NL explanation
E · 5 Specialist Agents (Nodes ② & ⑤)
🤖 Resume Parsing
Heuristic · zero LLM
Section detection
+ bias flags
(runs at Node ②)
🤖 Skill Matching
gpt-4o-mini
Skill coverage
matched/missing/partial
🤖 Experience
gpt-4o-mini
Years & seniority
Career progression
🤖 Technical Depth
gpt-4o-mini
Project complexity
Tech stack extraction
🤖 Culture Fit
gpt-4o ← stronger model
Soft skills assessment
LLM-as-judge role
G · API & Frontend
⚡ FastAPI Backend :8000
api/main.py + routes.py
POST /match
GET /categories
GET /resume/{id}
POST /feedback
GET /schedule
GET /handoffs
⚛️ React Frontend :3000
Vite + React Router v6
SearchPage
CandidatePage
AnalyticsPage
RecruiterPage
Guardrail UI (amber block)
traces
H · Feedback Loop & Observability
📊 Recruiter Feedback
Helpful/not helpful ratings
Interview scheduling
Handoff notes
SQLite analytics store
🔭 LangSmith Tracing
All pipeline nodes traced
Token count & latency
LANGCHAIN_TRACING_V2=true
Project: ai-resume-matching
🧪 DeepEval Tests
AnswerRelevancyMetric
GEval (LLM-as-judge)
RUN_LLM_JUDGE=1 flag
⚙️ Config
gating_thresholds.json
5 weight profiles
technical/analytics
management/biz_dev/general